

CAMP-MPPI: Contact-Aware Manifold Projection MPPI for Hard Dynamics-Level Contact-Wrench Constraints

Abstract—Contact-rich manipulation requires not only kinematic feasibility but also dynamics-level admissibility of interaction wrenches to prevent slip, contact loss, or object instability. Existing hard-constrained Model Predictive Path Integral (MPPI) methods primarily enforce kinematic feasibility, whereas force-aware MPPI formulations typically account for interaction forces through rollout objectives or soft penalties. This paper presents Contact-Aware Manifold Projection MPPI (CAMP-MPPI), a sampling-based control framework that jointly enforces contact-manifold consistency, kinematic inequalities, and hard dynamics-level wrench constraints within MPPI rollouts. CAMP-MPPI maps heterogeneous physical constraints into a common commanded-acceleration space, composes them into a single admissible set, and projects each sampled control to its closest feasible acceleration under an inertia-weighted metric. When a contact manifold is present, the projected controls are propagated through manifold-consistent dynamics. Experiments on a Franka manipulator demonstrate stable surface interaction while maintaining admissible contact wrenches and stable passive-object transport under support-wrench constraints. The results show that CAMP-MPPI provides a unified mechanism for enforcing coupled kinematic and dynamics-level feasibility directly within the MPPI sampling process. Project page: <https://anonymous.4open.science/w/camp-mppi/>.

I. INTRODUCTION

Many robotic manipulation tasks are constrained not only by where the robot moves, but also by how it physically interacts with the environment [1], [2]. In surface processing or writing, for example, the end-effector must remain consistent with the contact geometry while sustaining forces compatible with frictional and unilateral contact constraints [3], [4]. Likewise, in object transport and grasp-based manipulation, kinematic feasibility alone does not guarantee physical realizability when the support or grasp wrench required for execution exceeds the admissible range [5], [6]. Contact-rich manipulation therefore requires simultaneous consideration of contact-manifold consistency and dynamics-level wrench admissibility.

Addressing such problems within a predictive control framework requires repeated evaluation of state-dependent feasibility over the prediction horizon while maintaining real-time computational tractability. Model Predictive Path Integral (MPPI) control [7] is well suited to this setting. Its forward-rollout formulation permits state-dependent constraint violations to be evaluated along sampled trajectories and incorporated into the trajectory cost. The rollout evaluations are highly parallelizable, supporting real-time processing of a large number of samples without requiring explicit gradients of the task objective [7], [8].

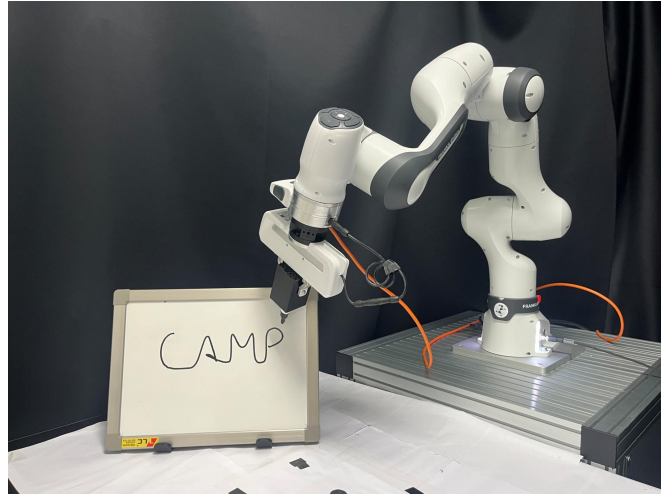


Fig. 1. Contact-aware surface writing with CAMP-MPPI, maintaining contact-manifold consistency and wrench feasibility throughout the motion.

Conventional MPPI formulations, however, typically encode state and input constraints through soft penalties in the rollout cost. Such penalties bias the sampling distribution toward desirable trajectories, whereas feasibility is not enforced as an intrinsic property of the sampled controls. Recent constrained MPPI methods instead modify the samples or rollouts themselves. PR-MPPI [9] confines rollouts to holonomic constraint manifolds, while MPPI-Tan [10], π -MPPI [11], CSC-MPPI [12], and MC-MPPI [13] respectively employ tangent-space sampling, feasible-set projection, infeasible-sample correction, or learned feasibility models. Control-barrier [14] and safety-filtering [15] variants similarly enforce state or input inequalities within the sampling loop. These approaches are effective for hard kinematic feasibility, but they do not directly represent whether the contact wrench required to realize a sampled motion is physically admissible.

MPPI-based manipulation has also been extended toward force-aware and contact-rich settings. Torque-sampling MPPI [16], [17] directly samples joint torques and propagates full rigid-body dynamics to generate compliant and force-aware behavior, while other approaches employ physics-based contact rollouts for non-prehensile manipulation or incorporate rollout contact forces into the objective [18], [19]. These studies demonstrate the potential of MPPI for contact-rich and force-aware manipulation, whereas contact-wrench admissibility is not explicitly constructed as a hard feasible set over the sampled control at each rollout stage. Moreover, the joint enforcement of

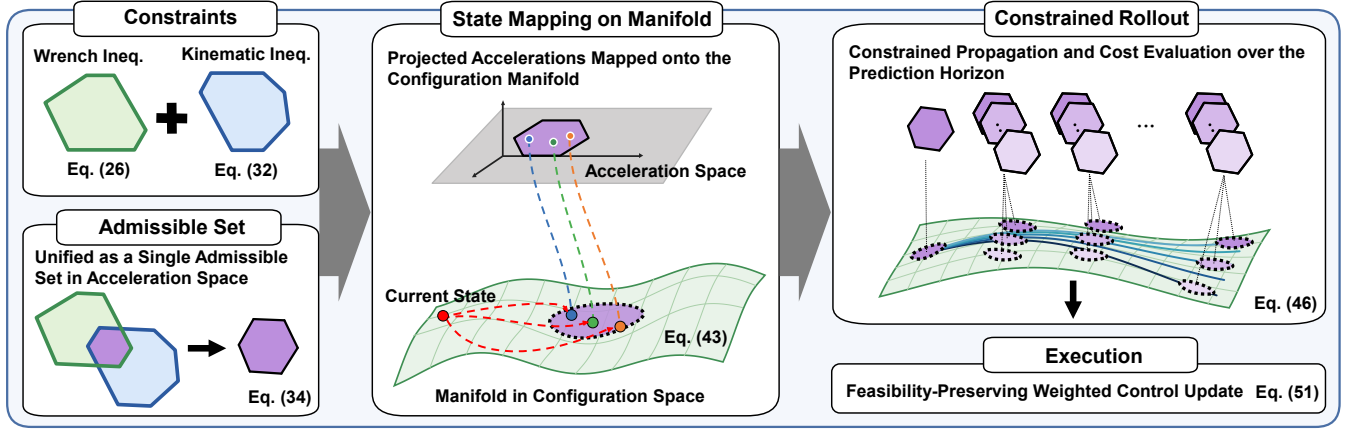


Fig. 2. Overview of the CAMP-MPPI control pipeline. Contact-wrench and kinematic constraints are mapped into a unified admissible set in commanded-acceleration space, onto which each sampled acceleration is jointly projected under the generalized inertia metric. The projected commands are propagated through manifold-constrained dynamics over the prediction horizon and evaluated by the task cost. A feasibility-preserving weighted update of the projected first-step commands yields the executed control, thereby enforcing dynamics-level wrench and kinematic feasibility directly within the MPPI sampling process.

dynamics-level wrench constraints and kinematic constraints within a single admissible control set remains largely unexplored.

This paper addresses this gap with Contact-Aware Manifold Projection MPPI (CAMP-MPPI). CAMP-MPPI maps heterogeneous physical constraints into a common commanded-acceleration space. At each rollout state, contact- and support-wrench admissibility are expressed directly as constraints on the sampled acceleration and combined with acceleration-level kinematic safety constraints. The resulting constraints define a single admissible set, onto which each raw MPPI proposal is projected under an inertia-weighted metric. The set is reconstructed at every rollout state, enabling CAMP-MPPI to enforce both dynamics-level wrench and kinematic feasibility as hard sample-wise admissibility constraints rather than through soft penalties. The main contributions of this work are:

- A dynamics-level wrench preimage formulation that maps contact- and support-wrench admissibility directly into hard constraints over the sampled commanded acceleration, enabling physical interaction feasibility to be enforced within MPPI rollouts.
- A unified composite projection framework that combines wrench-induced and kinematic inequalities in a common commanded-acceleration space and jointly projects each MPPI sample under the generalized inertia metric.
- A feasibility-preserving MPPI execution scheme that retains hard inequality admissibility after the weighted update at the executed step, while manifold consistency is maintained through constrained rollout dynamics.
- Real-world validation on surface writing and passive-object transport demonstrates simultaneous wrench and kinematic feasibility and improved task-level success over soft and partially constrained baselines, while maintaining real-time GPU execution.

II. MPPI PRELIMINARIES

MPPI optimizes a finite-horizon control sequence by sampling perturbations around a nominal sequence and reweighting the resulting rollouts according to their trajectory costs. Let $\mathbf{U} = (\mathbf{u}_0, \dots, \mathbf{u}_{H-1})$ denote the nominal control sequence over a horizon of length H . Given the current state, the cost associated with a rollout generated by $\mathbf{U}$ is

$$J(\mathbf{U}) = \sum_{t=0}^{H-1} \ell(\mathbf{x}_t, \mathbf{u}_t) + \phi(\mathbf{x}_H), \quad (1)$$

where $\ell(\cdot)$ and $\phi(\cdot)$ denote the running and terminal costs, respectively. MPPI reweights the base control distribution according to the trajectory cost,

$$q^*(\mathbf{U}) \propto \exp\left(-\frac{1}{\beta} J(\mathbf{U})\right) p(\mathbf{U}), \quad (2)$$

where $p(\mathbf{U})$ denotes the base control distribution and $\beta > 0$ is the temperature parameter. The distribution is approximated by sampling perturbation sequences around the current nominal sequence,

$$\mathbf{U}^k = \mathbf{U} + \mathbf{E}^k, \quad \mathbf{E}^k = (\epsilon_0^k, \dots, \epsilon_{H-1}^k), \quad (3)$$

where each time-step perturbation is independently sampled as

$$\epsilon_t^k \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad t = 0, \dots, H-1, \quad (4)$$

and Σ denotes the sampling covariance. After each sampled sequence is propagated through the rollout model, its trajectory cost J_k is converted into the normalized weight

$$w_k = \frac{\exp(-J_k/\beta)}{\sum_{r=1}^K \exp(-J_r/\beta)}. \quad (5)$$

The nominal sequence is then updated using the cost-weighted perturbation sequences,

$$\mathbf{U} \leftarrow \mathbf{U} + \sum_{k=1}^K w_k \mathbf{E}^k. \quad (6)$$

Thus, standard MPPI improves the nominal sequence by assigning larger weights to lower-cost rollouts. However, this cost-based weighting mechanism does not by itself guarantee that each sampled control satisfies hard constraints. This limitation is especially relevant in contact-rich manipulation, where preserving physical feasibility throughout the rollout is essential.

III. METHODOLOGY

In this section, we present CAMP-MPPI, which unifies dynamics-level wrench constraints and kinematic inequalities into a single admissible set in commanded-acceleration space. Each MPPI sample is projected onto this set, and, when a contact manifold is present, the projected command is propagated through manifold-consistent dynamics. The overall framework is illustrated in Fig. 2.

A. Problem Formulation

Consider a robotic system described by generalized coordinates $\mathbf{q} \in \mathbb{R}^n$ and generalized velocity $\dot{\mathbf{q}} \in \mathbb{R}^n$. CAMP-MPPI adopts the commanded generalized acceleration $\mathbf{a} \in \mathbb{R}^n$ as the sampled control variable. When persistent contact or a motion-level equality constraint is prescribed, the state must satisfy

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}, \quad \mathbf{J}_c(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0}. \quad (7)$$

The set of equality-consistent states is therefore defined as the phase-space constraint manifold

$$\bar{\mathcal{M}} = \{(\mathbf{q}, \dot{\mathbf{q}}) \mid \mathbf{c}(\mathbf{q}) = \mathbf{0}, \mathbf{J}_c(\mathbf{q})\dot{\mathbf{q}} = \mathbf{0}\}. \quad (8)$$

Maintaining this consistency during motion further requires the acceleration-level condition

$$\mathbf{J}_c(\mathbf{q})\ddot{\mathbf{q}} + \dot{\mathbf{J}}_c(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} = \mathbf{0}. \quad (9)$$

Contact-rich motion is additionally subject to dynamics-level wrench admissibility. Let λ_i denote the wrench associated with the i th contact, support, or grasp interaction, and let $\mathcal{K}_i$ denote its admissible wrench set. The corresponding physical feasibility condition is

$$\lambda_i \in \mathcal{K}_i, \quad i = 1, \dots, n_\lambda. \quad (10)$$

We also consider configuration-dependent kinematic constraints represented by safety functions $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$h_j(\mathbf{q}) \geq 0, \quad j = 1, \dots, n_h. \quad (11)$$

Let $\mathbf{A} = (\mathbf{a}_0, \dots, \mathbf{a}_{H-1})$ denote the commanded-acceleration sequence over a horizon of length H . CAMP-MPPI considers the constrained finite-horizon problem

$$\begin{aligned} \min_{\mathbf{A}} \quad & \mathbb{E} \left[\phi(\mathbf{x}_H) + \sum_{t=0}^{H-1} \ell(\mathbf{x}_t, \mathbf{a}_t) \right] \\ \text{s.t.} \quad & (\mathbf{q}_t, \dot{\mathbf{q}}_t) \in \bar{\mathcal{M}}, \\ & \lambda_{i,t} \in \mathcal{K}_i, \quad i = 1, \dots, n_\lambda, \\ & h_j(\mathbf{q}_t) \geq 0, \quad j = 1, \dots, n_h. \end{aligned} \quad (12)$$

Each CAMP-MPPI rollout maintains manifold consistency while satisfying the associated dynamics-level wrench and

kinematic constraints. To enforce these coupled requirements within a common formulation, we express both wrench and kinematic constraints directly in terms of the commanded acceleration under the manifold-constrained dynamics. The following subsections derive this representation and construct the resulting composite admissible set.

B. Dynamics-Level Wrench Preimage

We first examine how a sampled commanded acceleration determines the reaction wrench required to maintain a prescribed manifold constraint. Consider the robot dynamics subject to the manifold constraint in (9),

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau} + \mathbf{J}_c^\top(\mathbf{q})\boldsymbol{\lambda}, \quad (13)$$

where $\mathbf{M}(\mathbf{q}) \succ 0$ is the generalized inertia matrix, $\mathbf{b}(\mathbf{q}, \dot{\mathbf{q}})$ collects the Coriolis, centrifugal, and gravitational terms, and $\boldsymbol{\lambda}$ denotes the reaction wrench required to maintain the constrained motion on the manifold. For a commanded acceleration $\mathbf{a}$, we consider the inverse-dynamics input

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q})\mathbf{a} + \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}}), \quad (14)$$

which yields the actual robot acceleration

$$\ddot{\mathbf{q}} = \mathbf{a} + \mathbf{M}^{-1}\mathbf{J}_c^\top\boldsymbol{\lambda}. \quad (15)$$

Substituting (15) into the acceleration-level manifold consistency condition in (9) gives

$$\mathbf{J}_c\mathbf{a} + \mathbf{J}_c\mathbf{M}^{-1}\mathbf{J}_c^\top\boldsymbol{\lambda} + \dot{\mathbf{J}}_c\dot{\mathbf{q}} = \mathbf{0}. \quad (16)$$

The matrix $\mathbf{J}_c\mathbf{M}^{-1}\mathbf{J}_c^\top$ maps a reaction wrench in constraint space to the corresponding acceleration response along the constrained directions. Its inverse therefore represents the effective inertia observed in constraint space, which we define as

$$\boldsymbol{\Lambda}_c = (\mathbf{J}_c\mathbf{M}^{-1}\mathbf{J}_c^\top)^{-1}. \quad (17)$$

The reaction wrench required for manifold consistency is then given by

$$\boldsymbol{\lambda}_{\text{req}}(\mathbf{a}) = -\boldsymbol{\Lambda}_c (\mathbf{J}_c\mathbf{a} + \dot{\mathbf{J}}_c\dot{\mathbf{q}}). \quad (18)$$

At a fixed rollout state, $\mathbf{J}_c$, $\dot{\mathbf{J}}_c$, $\mathbf{M}$, and $\boldsymbol{\Lambda}_c$ are fixed quantities, and the required reaction wrench is affine in the commanded acceleration,

$$\boldsymbol{\lambda}_{\text{req}}(\mathbf{a}) = \mathbf{G}\mathbf{a} + \mathbf{g}, \quad (19)$$

where

$$\mathbf{G} = -\boldsymbol{\Lambda}_c\mathbf{J}_c, \quad \mathbf{g} = -\boldsymbol{\Lambda}_c\dot{\mathbf{J}}_c\dot{\mathbf{q}}. \quad (20)$$

More generally, for the i th interaction, the corresponding wrench is expressed as an affine function of the commanded acceleration,

$$\lambda_i(\mathbf{a}) = \mathbf{G}_i\mathbf{a} + \mathbf{g}_i, \quad (21)$$

where $\mathbf{G}_i$ and $\mathbf{g}_i$ depend on the corresponding interaction model and the current rollout state. Let the admissible wrench set for the i th interaction be

$$\mathcal{K}_i = \{\lambda_i \mid \mathbf{A}_i\lambda_i \leq \mathbf{b}_i\}. \quad (22)$$

Dynamics-level feasibility requires

$$\lambda_i(\mathbf{a}) \in \mathcal{K}_i. \quad (23)$$

Accordingly, the set of commanded accelerations that induce admissible interaction wrenches is defined as

$$\mathcal{A}_i^\lambda = \{\mathbf{a} \mid \lambda_i(\mathbf{a}) \in \mathcal{K}_i\}, \quad (24)$$

which is the preimage of $\mathcal{K}_i$ under the acceleration-to-wrench map. Substituting the affine wrench relation into the admissibility condition gives

$$\mathbf{A}_i(\mathbf{G}_i \mathbf{a} + \mathbf{g}_i) \leq \mathbf{b}_i, \quad (25)$$

which can be written compactly as

$$\mathbf{C}_i^\lambda \mathbf{a} \leq \mathbf{d}_i^\lambda, \quad (26)$$

where

$$\mathbf{C}_i^\lambda = \mathbf{A}_i \mathbf{G}_i, \quad \mathbf{d}_i^\lambda = \mathbf{b}_i - \mathbf{A}_i \mathbf{g}_i. \quad (27)$$

This acceleration-space formulation directly represents hard physical constraints originally defined in wrench space as feasibility conditions on the commanded acceleration. This provides a common representation for dynamics-level interaction constraints, which are combined with the kinematic constraints in Sec. III-C.

C. Composite Admissibility Projection

In this section, the kinematic constraints in (11) are mapped to commanded-acceleration space and combined with the wrench-induced inequalities to define a unified admissible set. For each safety function $h_j(\mathbf{q}) \geq 0$, we impose the second-order condition [20]

$$\ddot{h}_j + (k_{1,j} + k_{2,j})\dot{h}_j + k_{1,j}k_{2,j}h_j \geq 0, \quad (28)$$

where $k_{1,j}, k_{2,j} > 0$ are positive gains that regulate the approach to the safety boundary. Since h_j depends only on the configuration,

$$\dot{h}_j = \nabla h_j^\top \dot{\mathbf{q}}, \quad \ddot{h}_j = \nabla h_j^\top \ddot{\mathbf{q}} + \dot{\nabla h}_j^\top \dot{\mathbf{q}}. \quad (29)$$

To express (28) in terms of the commanded acceleration, we substitute the required reaction wrench in (18) into the constrained acceleration in (15), yielding

$$\ddot{\mathbf{q}}(\mathbf{a}) = \mathbf{F} \mathbf{a} + \mathbf{f}, \quad (30)$$

where

$$\mathbf{F} = \mathbf{I} - \mathbf{M}^{-1} \mathbf{J}_c^\top \mathbf{\Lambda}_c \mathbf{J}_c, \quad \mathbf{f} = -\mathbf{M}^{-1} \mathbf{J}_c^\top \mathbf{\Lambda}_c \dot{\mathbf{J}}_c \dot{\mathbf{q}}. \quad (31)$$

Combining the safety-function derivatives in (29) with the affine constrained dynamics in (30), the second-order condition in (28) reduces to an affine inequality in the commanded acceleration,

$$\mathbf{C}_j^h \mathbf{a} \leq \mathbf{d}_j^h. \quad (32)$$

Here, $\mathbf{d}_j^h$ includes a control-interval safety margin to account for barrier variation between consecutive control updates [21].

The wrench-induced inequalities derived in Sec. III-B and the kinematic inequalities above are stacked as

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_1^\lambda \\ \vdots \\ \mathbf{C}_{n_\lambda}^\lambda \\ \mathbf{C}_1^h \\ \vdots \\ \mathbf{C}_{n_h}^h \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} \mathbf{d}_1^\lambda \\ \vdots \\ \mathbf{d}_{n_\lambda}^\lambda \\ \mathbf{d}_1^h \\ \vdots \\ \mathbf{d}_{n_h}^h \end{bmatrix}, \quad (33)$$

which defines the composite admissible set

$$\mathcal{A}(\mathbf{q}, \dot{\mathbf{q}}) = \{\mathbf{a} \in \mathbb{R}^n \mid \mathbf{C} \mathbf{a} \leq \mathbf{d}\}. \quad (34)$$

CAMP-MPPI measures acceleration corrections under the generalized inertia metric. For the affine wrench variation

$$\Delta \lambda = \mathbf{G} \Delta \mathbf{a}, \quad (35)$$

the minimum- $\mathbf{M}$ -norm acceleration correction that produces a prescribed $\Delta \lambda$ satisfies

$$\|\Delta \mathbf{a}_{\min}\|_{\mathbf{M}}^2 = \Delta \lambda^\top (\mathbf{G} \mathbf{M}^{-1} \mathbf{G}^\top)^{-1} \Delta \lambda. \quad (36)$$

Thus, the inertia-weighted norm of the minimum acceleration correction corresponds to the induced weighted norm of the associated wrench variation. Accordingly, each raw sample is projected onto the composite admissible set as

$$\mathbf{a}^* = \arg \min_{\mathbf{a}} \frac{1}{2} (\mathbf{a} - \mathbf{a}^{\text{raw}})^\top \mathbf{M} (\mathbf{a} - \mathbf{a}^{\text{raw}}) \quad (37)$$

s.t. $\mathbf{C} \mathbf{a} \leq \mathbf{d}.$

The wrench and kinematic constraints are jointly enforced within a single admissible-set projection rather than through sequential corrections. Whenever $\mathcal{A}(\mathbf{q}, \dot{\mathbf{q}})$ is nonempty, the resulting $\mathbf{a}^*$ satisfies all included inequalities while minimally modifying the raw MPPI sample under the inertia metric. The projected command is then propagated through the manifold-constrained rollout dynamics of Sec. III-D to determine the corresponding reaction wrench and state acceleration.

D. CAMP-MPPI Rollout and Feasibility-Preserving Update

At each control cycle, CAMP-MPPI samples K commanded-acceleration sequences around a nominal sequence

$$\mathbf{A}^{\text{nom}} = (\mathbf{a}_0^{\text{nom}}, \dots, \mathbf{a}_{H-1}^{\text{nom}}). \quad (38)$$

The t th raw command of the k th sample is given by

$$\mathbf{a}_t^{k, \text{raw}} = \mathbf{a}_t^{\text{nom}} + \boldsymbol{\epsilon}_t^k, \quad \boldsymbol{\epsilon}_t^k \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t), \quad k = 1, \dots, K. \quad (39)$$

Each sample is independently propagated over the prediction horizon. At rollout step t , the current sampled state $(\mathbf{q}_t^k, \dot{\mathbf{q}}_t^k)$ determines the state-dependent composite admissible set

$$\mathcal{A}_t^k = \mathcal{A}(\mathbf{q}_t^k, \dot{\mathbf{q}}_t^k), \quad (40)$$

and the composite projection in (37) is applied to the raw command to obtain an admissible commanded acceleration satisfying

$$\mathbf{a}_t^{k, \star} \in \mathcal{A}_t^k. \quad (41)$$

The resulting admissible command is then propagated through the manifold-constrained dynamics. The reaction wrench required to maintain the manifold is

$$\lambda_{\text{req},t}^k = -\Lambda_{c,t}^k \left(\mathbf{J}_{c,t}^k \mathbf{a}_t^{k,*} + \dot{\mathbf{J}}_{c,t}^k \dot{\mathbf{q}}_t^k \right), \quad (42)$$

and the corresponding constrained state acceleration is

$$\ddot{\mathbf{q}}_t^k = \mathbf{a}_t^{k,*} + (\mathbf{M}_t^k)^{-1} (\mathbf{J}_{c,t}^k)^\top \lambda_{\text{req},t}^k. \quad (43)$$

Consequently,

$$\mathbf{J}_{c,t}^k \ddot{\mathbf{q}}_t^k + \dot{\mathbf{J}}_{c,t}^k \dot{\mathbf{q}}_t^k = \mathbf{0}, \quad (44)$$

so each rollout maintains manifold consistency through the constrained dynamics while satisfying the composite wrench and kinematic admissibility constraints through the projected command.

The rollout state is propagated using the constrained acceleration,

$$\dot{\mathbf{q}}_{t+1}^k = \dot{\mathbf{q}}_t^k + \Delta t \ddot{\mathbf{q}}_t^k, \quad \mathbf{q}_{t+1}^k = \mathbf{q}_t^k + \Delta t \dot{\mathbf{q}}_{t+1}^k. \quad (45)$$

At the next rollout step, the admissible set is reconstructed from the state obtained by propagating the previously projected command. Thus, projection and manifold-constrained propagation are repeated sequentially over the entire prediction horizon. Repeating this procedure over the prediction horizon yields the full-horizon rollout cost

$$J_k = \phi(\mathbf{x}_H^k) + \sum_{t=0}^{H-1} \ell(\mathbf{x}_t^k, \mathbf{a}_t^{k,*}), \quad (46)$$

from which the normalized MPPI weights are computed as

$$w_k = \frac{\exp(-J_k/\beta)}{\sum_{r=1}^K \exp(-J_r/\beta)}, \quad w_k \geq 0, \quad \sum_{k=1}^K w_k = 1. \quad (47)$$

The nominal sequence is updated using the weighted projected commands,

$$\mathbf{a}_t^{\text{nom}} \leftarrow \sum_{k=1}^K w_k \mathbf{a}_t^{k,*}, \quad t = 0, \dots, H-1. \quad (48)$$

For $t > 0$, the resulting future nominal sequence serves as a cost-weighted reference that summarizes the future control preferences of low-cost constrained rollouts and guides subsequent sampling. However, the sampled rollouts generally evolve to different states $(\mathbf{q}_t^k, \dot{\mathbf{q}}_t^k)$, resulting in different state-dependent admissible sets $\mathcal{A}_t^k$. Therefore, although each projected command satisfies $\mathbf{a}_t^{k,*} \in \mathcal{A}_t^k$, their weighted average is not generally guaranteed to be feasible. In contrast, at $t = 0$, all samples originate from the same measured state $(\mathbf{q}_0, \dot{\mathbf{q}}_0)$ and therefore share the same composite admissible set,

$$\mathcal{A}_0^1 = \dots = \mathcal{A}_0^K = \mathcal{A}_0. \quad (49)$$

Hence, every projected first-step sample satisfies

$$\mathbf{a}_0^{k,*} \in \mathcal{A}_0, \quad \mathbf{C}_0 \mathbf{a}_0^{k,*} \leq \mathbf{d}_0. \quad (50)$$

CAMP-MPPI applies the first element of the weighted projected sequence as the executed command,

$$\mathbf{a}_0^{\text{exec}} = \sum_{k=1}^K w_k \mathbf{a}_0^{k,*}. \quad (51)$$

Since the composite admissible set

$$\mathcal{A}_0 = \{\mathbf{a} \mid \mathbf{C}_0 \mathbf{a} \leq \mathbf{d}_0\} \quad (52)$$

is convex, the weighted command satisfies

$$\mathbf{C}_0 \mathbf{a}_0^{\text{exec}} = \sum_{k=1}^K w_k \mathbf{C}_0 \mathbf{a}_0^{k,*} \leq \sum_{k=1}^K w_k \mathbf{d}_0 = \mathbf{d}_0, \quad (53)$$

which implies

$$\mathbf{a}_0^{\text{exec}} \in \mathcal{A}_0. \quad (54)$$

The executed command incorporates the full-horizon rollout preference encoded by the MPPI weights while preserving first-step feasibility through convex averaging over the common admissible set.

Applying the executed command through the manifold-constrained dynamics gives

$$\lambda_{\text{req}}(\mathbf{a}_0^{\text{exec}}) = -\Lambda_c \left(\mathbf{J}_c \mathbf{a}_0^{\text{exec}} + \dot{\mathbf{J}}_c \dot{\mathbf{q}}_0 \right), \quad (55)$$

and

$$\ddot{\mathbf{q}}_0 = \mathbf{a}_0^{\text{exec}} + \mathbf{M}^{-1} \mathbf{J}_c^\top \lambda_{\text{req}}(\mathbf{a}_0^{\text{exec}}), \quad (56)$$

which satisfies

$$\mathbf{J}_c \ddot{\mathbf{q}}_0 + \dot{\mathbf{J}}_c \dot{\mathbf{q}}_0 = \mathbf{0}. \quad (57)$$

Inequality admissibility is preserved after MPPI weighted averaging through $\mathbf{a}_0^{\text{exec}} \in \mathcal{A}_0$, while equality consistency is maintained by the manifold-constrained state acceleration generated from the same command. At the next control cycle, the composite admissible set is reconstructed from the newly measured state and the horizon optimization is repeated.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate CAMP-MPPI on two real-world manipulation tasks involving dynamics-level wrench constraints: curved-surface writing under persistent contact and tray transport in an obstacle-constrained environment. The former combines contact-manifold consistency with contact-wrench feasibility, while the latter combines kinematic collision avoidance with support-wrench feasibility. For both tasks, CAMP-MPPI is compared against Vanilla MPPI [7], PR-MPPI [9], and Torque-Sampling MPPI [16]. Vanilla MPPI uses standard cost-based constraint handling, PR-MPPI enforces manifold and kinematic constraints through projection-based hard constraints, and Torque-Sampling MPPI incorporates wrench-related objectives as soft rollout costs.

All experiments are conducted on a Franka Research 3 (FR3) manipulator with MPPI rollouts evaluated in parallel on an NVIDIA GeForce RTX 4070 GPU. We use $K = 1024$ samples for both tasks, with prediction horizons of $H = 15$ for surface writing and $H = 36$ for tray transport. Each

TABLE I
PERFORMANCE AND CONTACT-LOAD FEASIBILITY ON THE CURVED-SURFACE WRITING TASK.

Method	Path RMSE [mm]	Manifold RMS [mm]	Load Feas.[%]	Max Viol.[N]	Completion [%]	Comp. Time [ms]
Vanilla MPPI [7] [†]	16.11 ± 14.11	1.675 ± 10.31	26.2 ± 7.1	27.1 ± 18.9	11 ± 0.6	3.02 ± 0.02
PR-MPPI [9] [†]	0.686 ± 0.109	0.590 ± 0.098	52.6 ± 5.6	7.4 ± 18.3	39 ± 1.8	3.18 ± 0.34
Torque-Sampling MPPI [16] [†]	2.249 ± 0.380	1.211 ± 0.720	88.4 ± 3.2	2.9 ± 1.6	84 ± 6.7	4.33 ± 0.03
CAMP-MPPI	0.735 ± 0.134	1.087 ± 0.011	100.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0	6.68 ± 0.11

[†] Metrics are computed over the executed portion only.

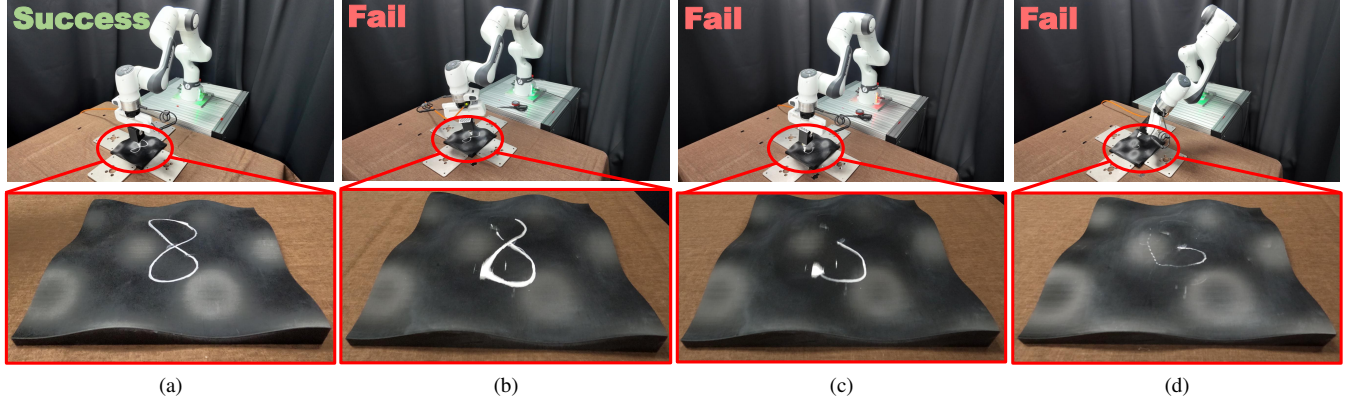


Fig. 3. Representative snapshots of the curved-surface writing task. CAMP-MPPI jointly satisfies the surface-contact manifold constraint and admissible contact-wrench constraints, enabling the robot to track the prescribed writing path while maintaining sustained contact with the curved surface. This experiment demonstrates the applicability of CAMP-MPPI to contact-rich manipulation requiring simultaneous geometric and force-level feasibility.

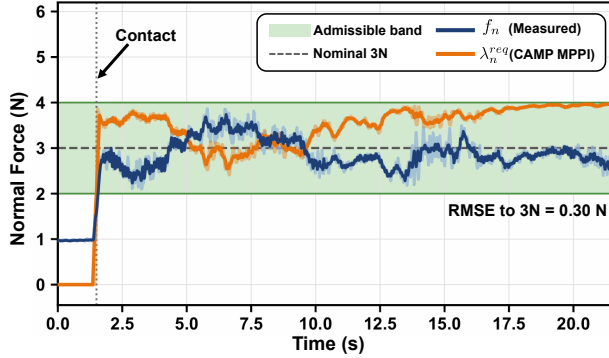


Fig. 4. Wrench admissibility and measured force tracking during curved-surface writing. After contact establishment, the required normal wrench computed by CAMP-MPPI remains within the prescribed admissible band, while the measured normal force tracks the nominal 3 N contact level.

method is evaluated over 10 trials for each task. CAMP-MPPI achieves mean computation times below 10 ms in both tasks, enabling 100 Hz real-time execution. Complete implementation details and experiment videos are provided on the project page.¹

B. Surface Writing under Contact-Manifold and Wrench Constraints

We first evaluate CAMP-MPPI on a curved-surface writing task requiring persistent contact. The robot follows a prescribed figure-eight path while maintaining contact-manifold consistency and keeping the required contact wrench within its admissible region. Fig. 3 compares representative writing outcomes, while Table I reports path tracking, manifold con-

sistency, wrench feasibility, maximum wrench violation, and task completion. CAMP-MPPI produces the most complete trace while maintaining wrench admissibility throughout the motion.

The baseline results reveal distinct failure modes associated with how the interaction constraints are handled. Torque-Sampling MPPI reproduces much of the prescribed path, but its soft treatment of wrench-related objectives permits contact loading to deviate from the admissible range, producing over-pressed regions together with faint or missing marks. PR-MPPI preserves geometric consistency with the surface, but the absence of a wrench-related term allows excessive contact loading, resulting in stamped or heavily pressed traces. Vanilla MPPI treats manifold consistency only through the rollout cost, leading to larger geometric deviation and incomplete path execution. These differences indicate that geometric consistency or force-aware cost shaping alone is insufficient to ensure physically admissible surface interaction.

Path RMSE and manifold RMS are computed only over the executed portion of each trial and should therefore be interpreted together with task completion. In particular, relatively small geometric errors from incomplete baselines do not imply superior full-task performance. Fig. 3 and Table I instead show that successful writing requires both accurate surface following and admissible contact loading. Fig. 4 further examines the interaction behavior of CAMP-MPPI after contact establishment. The required normal wrench remains within the prescribed admissible band, while the measured normal force tracks the nominal 3 N contact level. This result confirms that CAMP-MPPI preserves the geometric contact

¹<https://anonymous.4open.science/w/camp-mppi/>

TABLE II
SUPPORT-WRENCH FEASIBILITY, TASK OUTCOMES, AND COMPUTATION TIME ON THE TRAY-TRANSPORT TASK.

Method	Wrench Feas. [%]	Max Viol. [mm]	Viol. Time [%]	Peak $ a_h /a_{\max}$ [%]	Task Outcomes [†] [Upright / Coll. / Tip.]	Comp. Time [ms]
Vanilla MPPI [7]	80.2 ± 2.4	41.1 ± 9.7	19.8 ± 4.0	137 ± 11.9	0 / 3 / 7	5.23 ± 0.33
PR-MPPI [9]	82.4 ± 4.0	96.7 ± 12.4	17.6 ± 4.3	160 ± 19.6	0 / 0 / 10	5.81 ± 0.31
Torque-Sampling MPPI [16]	86.2 ± 6.2	35.1 ± 13.5	18.8 ± 6.2	125 ± 26.1	0 / 8 / 2	6.76 ± 0.58
CAMP-MPPI	100.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	97 ± 1.9	10 / 0 / 0	8.25 ± 0.39

[†]Task outcomes report the number of upright completions, collisions, and tipping failures.



Fig. 5. Representative snapshots of the tray-transport task. CAMP-MPPI jointly satisfies kinematic collision-avoidance and dynamics-level support-wrench constraints, enabling the robot to avoid obstacles while preventing excessive support wrenches that could tip the passive object. This experiment demonstrates the applicability of CAMP-MPPI to tasks governed solely by heterogeneous inequality constraints.

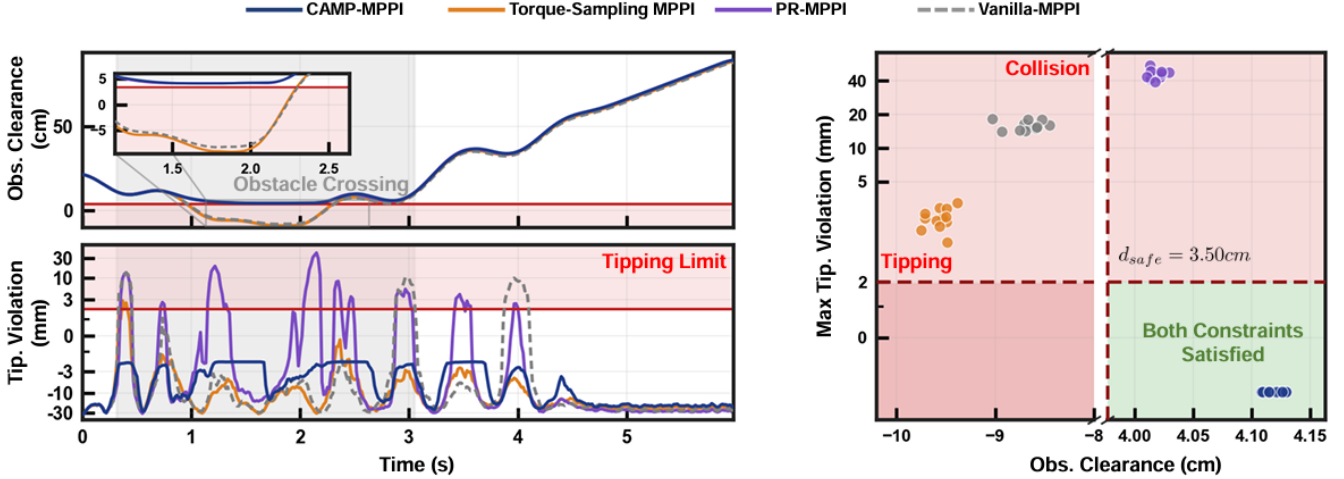


Fig. 6. Constraint satisfaction in the tray-transport task. The left plots report the minimum obstacle clearance and tipping-violation over time, with the red boundaries indicating the prescribed admissible limits and the shaded regions denoting constraint violation. The right plot summarizes the trial-level relationship between minimum obstacle clearance and maximum tipping violation. CAMP-MPPI remains within the region satisfying both constraints across trials, whereas the baselines violate the collision-avoidance constraint, the tipping constraint, or both.

constraint while maintaining physically admissible contact loading throughout the writing task.

C. Tray Transport under Heterogeneous Inequality Constraints

We next evaluate CAMP-MPPI on a tray-transport task in which a tall passive object is carried through an obstacle-constrained environment. Feasibility is jointly determined by a kinematic collision-avoidance constraint and a dynamics-level support-wrench constraint. The robot must maintain sufficient obstacle clearance while regulating the tray motion so that the support wrench required to stabilize the passive object remains within its admissible region. Fig. 5 compares representative outcomes of the four methods, while Table II summarizes wrench feasibility, maximum violation, constraint-violation time, peak acceleration, and the final

object state. CAMP-MPPI is the only method that satisfies both constraints while avoiding the obstacle and keeping the object upright.

The baseline results exhibit distinct failure modes associated with incomplete treatment of the coupled constraints. Torque-Sampling MPPI captures the force-aware interaction through its rollout cost, but the soft treatment of wrench-related objectives permits a trade-off between support-wrench regulation and obstacle avoidance. As a result, the motion either becomes insufficiently evasive or induces excessive tray acceleration, leading to collision or loss of object stability. PR-MPPI reliably preserves the kinematic collision-avoidance constraint, but the absence of a support-wrench term allows aggressive evasive motion that violates the tipping-related constraint and destabilizes the

passive object. Vanilla MPPI provides no hard feasibility mechanism for either requirement, resulting in inconsistent obstacle avoidance and object stability.

Fig. 6 further highlights these differences in constraint space. CAMP-MPPI maintains the prescribed obstacle-clearance margin while keeping the tipping violation within its admissible limit. The trial-level scatter plot shows the same separation, with CAMP-MPPI concentrated in the region where both constraints are simultaneously satisfied, whereas the baselines violate the collision-avoidance constraint, the tipping constraint, or both. These results show that handling the heterogeneous inequality constraints separately is insufficient for task-level feasibility. By enforcing them within a single composite admissible set, CAMP-MPPI retains the maneuverability required for obstacle avoidance while suppressing support-wrench conditions that would destabilize the transported object.

V. CONCLUSION

This paper presented CAMP-MPPI, a constrained sampling-based control framework that incorporates dynamics-level wrench admissibility directly into MPPI rollouts. Contact-wrench and kinematic constraints are mapped into a unified admissible set in commanded-acceleration space, and each sample is projected under the generalized inertia metric while manifold consistency is maintained through constrained dynamics. Real-world experiments on curved-surface writing and tray transport demonstrated that CAMP-MPPI maintains wrench feasibility together with geometric or kinematic constraints, whereas soft or partial constraint handling leads to reduced task completion and physical instability. These results highlight the importance of enforcing dynamics-level wrench admissibility directly within the MPPI sampling process for contact-rich manipulation. Future work will address contact-mode transitions, uncertainty-aware wrench constraints, and more general multi-contact interactions.

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